SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20% **QUESTIONS:** 05 **Total Marks:** 50

Annexure A

COMPARATIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth (2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total (10)
Weightage	20%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:
I **Manoj** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***Manoj***

RUBRICS FOR CALCULATION: Comparative Analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1. Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration: 45 Minutes   Assessment Tool 3 (Low)

Q1. Dataset: Customer Data

Customer	Income (Rs.)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question: Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 - 1 Mark)

Answer:

On this small 5-customer dataset (Income vs Spending Score), K-means requires the number of clusters (k) to be fixed in advance, e.g. k=2 separating high-income/low-spending customers (C2, C5) from low-income/high-spending customers (C1, C3, C4). It assigns points to the nearest centroid and iteratively updates centroids, converging quickly since the data is numeric and small. Hierarchical clustering instead builds a dendrogram by successively merging the closest pairs (e.g., C1 and C4 merge first due to similar Income/Spending values), letting the analyst choose the number of clusters after seeing the tree, without deciding k beforehand.

Performance: K-means gives a lower reconstruction (within-cluster) error faster because it directly optimizes centroid distances, whereas hierarchical clustering does not optimize a global objective and can be more sensitive to a single unusual merge.

Scalability: K-means has roughly O(n*k*i) complexity (n=points, k=clusters, i=iterations), which scales well as data grows. Hierarchical clustering has O(n^2 log n) or worse complexity because it must repeatedly compute and update the distance matrix between all clusters, making it impractical for large customer databases even though it works fine here for 5 records.

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question: Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 - 1 Mark)

Answer:

In this document-term dataset, D1, D2 and D5 have fairly similar word-frequency profiles (moderate Word2/Word3 weights) and could be seen as an overlapping group, while D3 and D4 form a clearly separate, distinct cluster with high Word1 weight. K-means performs hard assignment: every document is forced into exactly one cluster based on nearest centroid (Euclidean distance), so a borderline case like D5 (0.2, 0.4, 0.4) is placed entirely in one cluster even though its values are close to the boundary between groups, losing information about its partial similarity to both. The EM algorithm (fitting a Gaussian Mixture Model) instead performs soft/probabilistic assignment: it computes the probability that each document belongs to each cluster, so D5 could be assigned e.g. 60% to the D1/D2 group and 40% to another, correctly reflecting the overlap. This makes EM more suitable than K-means when clusters are not clearly separated, at the cost of higher computational complexity and sensitivity to initialization.

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question: Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 - 1 Mark)

Answer:

For these latitude/longitude points, Euclidean distance measures the straight-line ('as the crow flies') separation between two points, e.g. between P1 (13.08, 80.27) and P2 (13.10, 80.25) it is sqrt((0.02)^2 + (0.02)^2). Manhattan distance instead sums the absolute differences along each axis (|latitude difference| + |longitude difference|), which corresponds to movement along a grid, similar to travelling along city blocks or road networks rather than in a straight line. Because Euclidean distance is smaller than or equal to Manhattan distance for the same pair of points, clusters formed using Euclidean distance tend to be more compact and circular, while Manhattan distance tends to produce more elongated, diamond-shaped clusters and can be more robust to outliers in one dimension (e.g., a point with an unusual longitude but normal latitude). For geographic data where actual travel follows road grids, Manhattan distance can better reflect real-world proximity, whereas Euclidean distance is preferred when true physical distance is the relevant measure, so the choice of metric directly changes which points end up grouped together.

Q4. Dataset: Retail Products

Product	Sales (Rs.)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question: Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 - 1 Mark)

Answer:

Agglomerative clustering is a bottom-up approach: it starts with each product (P1-P5) as its own cluster and repeatedly merges the two closest clusters, so P1 and P3 (both high sales ~50000-52000, similar frequency 20-22) would merge early, followed by P2 and P5 (moderate sales/frequency), building up to larger groups. Divisive clustering is top-down: it starts with all five products in one cluster and recursively splits the least cohesive cluster, for example first separating the high-sales pair (P1, P3) from the rest, then splitting further. For this small retail dataset of only 5 products, agglomerative clustering is more suitable and is far more commonly used in practice, since computing all pairwise merges is cheap at this scale and gives a clear dendrogram of which products behave similarly. Divisive clustering is computationally more expensive (it must evaluate all possible ways to split a cluster) and only becomes advantageous when a coarse, high-level split of a very large product catalog is needed before drilling into finer groups.

Q5. Dataset: Mixed Attributes

Item	Price (Rs.)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question: Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 - 1 Mark)

Answer:

Centroid-based clustering (e.g., K-means) represents each cluster by the mean of its members and requires purely numeric attributes, since a 'mean' cannot be computed directly on a categorical attribute like Category (A/B). Applied here, Category would need to be encoded numerically first, and K-means would tend to group I1, I3, I5 (Category A, higher Price ~500-550, higher Rating 4.5-4.7) as one cluster and I2, I4 (Category B, lower Price and Rating) as another, assuming roughly spherical, similarly sized clusters. Density-based clustering (e.g., DBSCAN) instead groups points that are closely packed together (high density) and marks isolated points as noise/outliers, without needing to pre-specify the number of clusters and without computing a mean, so it can more naturally handle non-numeric or mixed attributes when combined with an appropriate distance measure (e.g., Gower distance) for Category. For this mixed-attribute dataset (numeric Price/Rating plus categorical Category), density-based clustering is generally more effective because it does not assume convex clusters or require averaging categorical values, while centroid-based clustering is simpler and faster but less accurate unless the categorical attribute is carefully encoded and the clusters are naturally round and evenly sized.